

Reconstructing frustrated signs from amplitudes with fixed-node and Krylov updates

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We separate the sign and positive-amplitude parts of a real many-body wave function, $\psi(x) = s(x)a(x)$ with $a(x) = |\psi(x)| > 0$, in the frustrated square-lattice spin-1/2 J_1 - J_2 Heisenberg model. For a fixed amplitude, one shifted Krylov step gives $s'(x) = s(x) \text{sgn}[T - r_s(x)]$, where $r_s(x) = (H\psi)(x)/\psi(x)$. At $J_2/J_1 = 0.5$ this reduces the exact 4×4 probability weight on incorrectly signed configurations from 1.27×10^{-2} to 3.41×10^{-4} . On 6×6 , using exactly the same neural-network amplitude, the reconstructed signs give the same energy as the original neural-network signs within sampling uncertainty. Conversely, lattice fixed-node projection improves the positive amplitude while keeping the signs fixed; an exact 4×4 fixed-node/Krylov loop reaches the exact ground-state signs. The reconstructed signs remain useful on 8×8 . The remaining scaling problem is to infer from fixed-node Monte Carlo walkers a positive amplitude function that can be evaluated on arbitrary configurations. Maximum likelihood targets $a_{\text{new}}^2 \propto a\phi_{\text{FN}}$, and stochastic reconfiguration produces an update that generalizes between independent walker samples.

I. INTRODUCTION

The Monte Carlo sign problem is severe in frustrated spin systems [1, 2]. We use the square-lattice J_1 - J_2 Heisenberg model as a benchmark. At $J_2 = 0$ its ground-state signs obey the Marshall rule [3]; near $J_2/J_1 = 1/2$ that rule fails and accurate variational states become nontrivial [4, 5].

We ask whether signs and positive amplitudes can be improved separately. For a real wave function we write

$$\psi(x) = s(x)a(x), \quad s(x) = \pm 1, \quad a(x) = |\psi(x)| > 0. \quad (1)$$

Given a , a single Krylov step updates the signs. Given s , lattice fixed node gives a sign-free positive-amplitude problem [6]. Our calculations first test these two operations separately and then close the feedback loop.

II. MODEL AND METHOD

We study

$$H = J_1 \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle ij \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (2)$$

on periodic $L \times L$ lattices with $J_1 = 1$ in the $S^z = 0$ sector. For a spin configuration x , the Marshall rule is $s_M(x) = (-1)^{N_{\downarrow}^A(x)}$ up to an overall sign, where $N_{\downarrow}^A$ counts down spins on one checkerboard sublattice.

A. One-step sign update

For $\psi = sa$, define the local energy

$$r_s(x) = \frac{(H\psi)(x)}{\psi(x)}. \quad (3)$$

Because

$$[(T - H)\psi](x) = [T - r_s(x)]\psi(x), \quad (4)$$

the sign of the shifted Krylov vector is

$$s_K(x) = s(x) \text{sgn}[T - r_s(x)]. \quad (5)$$

Only this sign is kept; the Krylov amplitude factor $|T - r_s(x)|$ is discarded. Thus the method is not repeated Lanczos multiplication. For exact systems, T is chosen to minimize the fixed-amplitude variational energy. For sampled systems, T is obtained from a two-cluster split of the $r_s(x)$ values and therefore does not use target sign labels.

When the exact ground state $\psi_0 = s_0 a_0$ is available, we fix the irrelevant overall sign and report directly the wrong-sign probability weight

$$w_s = \sum_{x: s(x) \neq s_0(x)} a_0(x)^2. \quad (6)$$

B. Fixed-node amplitude update

For a trial wave function ψ_T , an off-diagonal matrix element connecting x and y is sign violating when $H_{xy}\psi_T(y)/\psi_T(x) > 0$. Lattice fixed node removes such off-diagonal terms and adds their contribution to the diagonal [6]. After importance sampling, the resulting Hamiltonian is sign free and keeps the signs of ψ_T fixed.

Let $\phi_{\text{FN}}(x) > 0$ be the positive ground-state amplitude obtained from this fixed-node Hamiltonian. In an exact small-system calculation, ϕ_{FN} can be inserted directly into Eq. (5). In a walker calculation, however, the stationary sampled distribution is

$$f(x) \propto a(x)\phi_{\text{FN}}(x). \quad (7)$$

If a positive model is trained by maximum likelihood with $p_\theta(x) \propto a_\theta(x)^2$, the exact maximum-likelihood target is

$$a_{\text{new}}^2(x) \propto a(x)\phi_{\text{FN}}(x). \quad (8)$$

Since $a_{\text{new}}(x) > 0$, this is equivalent to taking the geometric mean of the current and fixed-node amplitudes.

The 4×4 exact calculations use the full $S^z = 0$ Hilbert space of 12870 states. For 6×6 and 8×8 we use a previously optimized translationally invariant vision-transformer neural quantum state (ViT) at $J_2/J_1 = 0.5$. In the sign-only benchmarks, its positive amplitude $|\psi_{\text{ViT}}(x)|$ is held fixed. The 8×8 fixed-node calculations use a walker population $M = 128$ and projection length $\beta = 1.2$.

III. RESULTS

A. One-step proof of concept

The simplest test keeps the positive amplitude fixed and changes only the signs. On 4×4 the exact ground state is available, so we can measure both the sign error and the corresponding energy error directly.

At $J_2/J_1 = 0.5$, the wrong-sign probability decreases from 1.27×10^{-2} to 3.41×10^{-4} , a factor 37.4. With the amplitude unchanged, the energy error per site falls from 1.16×10^{-2} to 5.51×10^{-4} , a factor 21.0. Thus one Krylov step removes about 97% of the wrong-sign probability and about 95% of the fixed-amplitude energy error.

B. Larger-system validation

We next remove the exact-sign benchmark. On 6×6 we use the same ViT positive amplitude with three sign choices: Marshall, one-step Krylov, and the original ViT signs.

The 6×6 energies per site are

$$\begin{aligned} e_{\text{Marshall}} &= -0.4981534, \\ e_{\text{Krylov}} &= -0.5033337, \\ e_{\text{ViT}} &= -0.5034504. \end{aligned} \quad (9)$$

The paired difference $e_{\text{Krylov}} - e_{\text{ViT}} = 1.17(287) \times 10^{-4}$ is unresolved, whereas Krylov improves over Marshall by 4.79×10^{-3} per site. The larger-system data therefore support the same conclusion as the exact benchmark: the one-step sign update captures the energetically relevant sign structure.

C. Closing the fixed-node–Krylov loop

We now test whether the positive amplitude can also be updated. We target $J_2/J_1 = 0.5$ but initialize with the exact $J_2 = 0$ positive amplitude and Marshall signs. Each iteration solves the current fixed-node problem exactly, uses its positive ground-state amplitude, and re-computes Eq. (5).

The wrong-sign weight falls from 1.27×10^{-2} initially to 2.03×10^{-3} after two iterations and 8.85×10^{-7} after twenty. At iteration 100 it is zero to numerical precision. The final amplitude fidelity is 0.9999946 and the trial-state energy is only 2.60×10^{-5} above the exact ground state.

D. Scaling to 8×8

At $M = 128$, the mean total fixed-node energies from two independent walker populations are

$$E_{\text{Krylov}} = -31.88542, \quad E_{\text{Marshall}} = -31.67748. \quad (10)$$

Using the Krylov signs gives $E/N = -0.498210$, compared with -0.494961 for Marshall signs. Figure 4 places this in the range of high-accuracy same-geometry benchmarks. With only two populations we treat their spread as a reproducibility scale rather than a precision error bar.

E. Learning the positive amplitude from walkers

For sampled fixed-node walkers, Eq. (8) defines the amplitude learning target. Its exact 4×4 behavior and the first scalable 8×8 learning step are shown in Fig. 5.

On 8×8 , ordinary maximum-likelihood gradients from two independent walker samples have Euclidean cosine -0.853 , so a standard gradient step does not transfer between samples. With stochastic reconfiguration,

$$\delta\theta = (S + \lambda I)^{-1} g_1, \quad (11)$$

and $\lambda = 1$, the directional derivative on the second sample is $+3.85$. A finite step $\eta = 0.01$ gives a positive independent-sample log-likelihood change of 0.0156 with effective sample size 3236/4096. This establishes a transferable first amplitude update, but not yet convergence of repeated 8×8 feedback iterations.

IV. DISCUSSION AND OUTLOOK

The data separate the two tasks. With a good positive amplitude, the dominant frustrated sign correction is captured by a threshold in the local energy. Fixed node solves the complementary positive-amplitude problem at fixed signs, and the exact 4×4 loop shows that the two updates can close.

The unresolved large-system problem is learning that positive amplitude from projector walkers. SR makes one 8×8 update generalize between independent walker samples, but repeated updates still require fixed-node energy validation. The method also requires a useful initial amplitude and should not be confused with repeated bare Krylov iteration; the direct second step is worse on 6×6 .

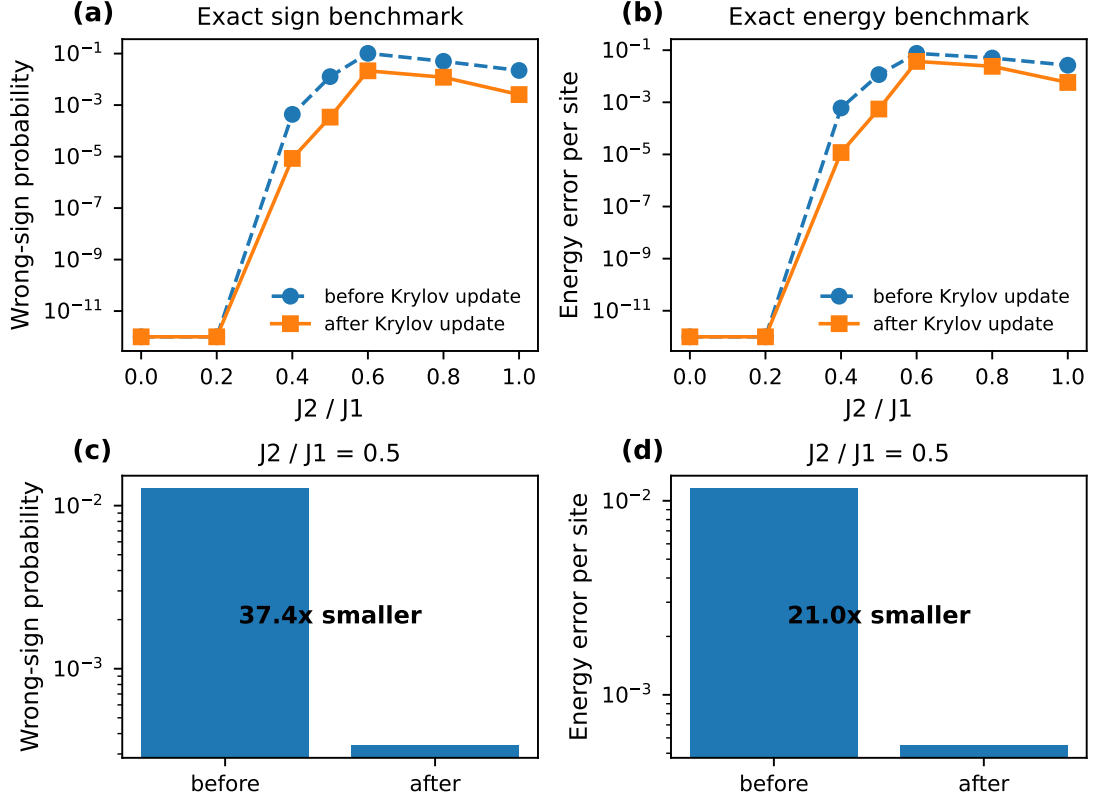


FIG. 1. One-step proof of concept on 4×4 . (a) Exact probability weight carried by configurations with the wrong sign. (b) Variational energy error per site using the same exact positive amplitude. (c,d) The same two quantities at $J_2/J_1 = 0.5$, shown explicitly before and after one Krylov sign update: the wrong-sign probability is reduced by a factor 37.4 and the energy error by a factor 21.0. For $J_2/J_1 \leq 0.6$ the initial sign rule is Marshall; for $J_2/J_1 \geq 0.8$ we use the corresponding stripe reference pattern. Exact zeros are shown at 10^{-12} on logarithmic axes.

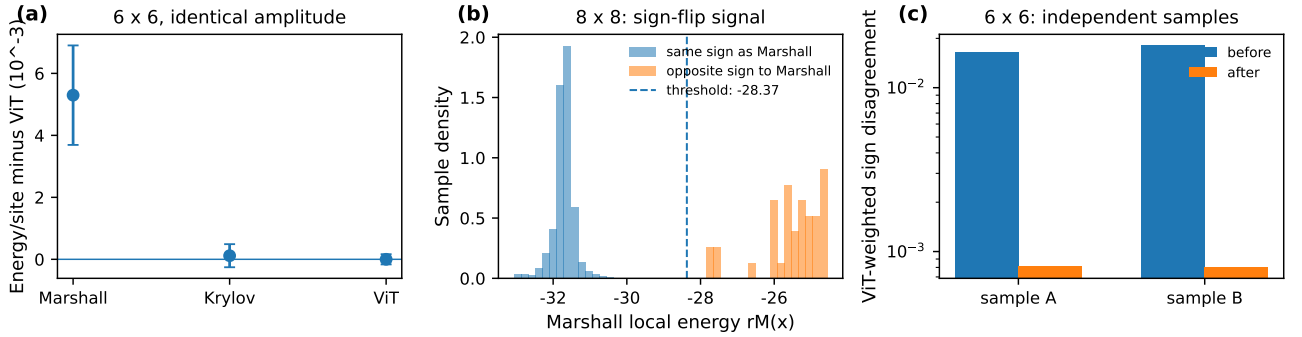


FIG. 2. Larger-system checks. (a) 6×6 energy with an identical ViT positive amplitude and three sign structures. (b) On 8×8 , the Marshall local energy separates configurations whose ViT sign agrees or disagrees with Marshall; the threshold is obtained from the local-energy data alone. (c) On two independent 6×6 samples, the fraction of $|\psi_{\text{ViT}}|^2$ weight whose sign disagrees with the ViT sign drops by about a factor of twenty after the Krylov update.

Appendix A: Direct second Krylov step

On 6×6 , retaining the full first Krylov amplitude gives $a_1(x) = |T_1 - r_0(x)|a_0(x)$ with coefficient of variation about 0.18. A second unsupervised threshold changes only 0.8–2.0% of the $|\psi_{\text{ViT}}|^2$ probability weight but raises

the fixed-amplitude energy by 0.65–0.96 in total energy. The feedback construction therefore restores or re-solves the amplitude before another sign update.

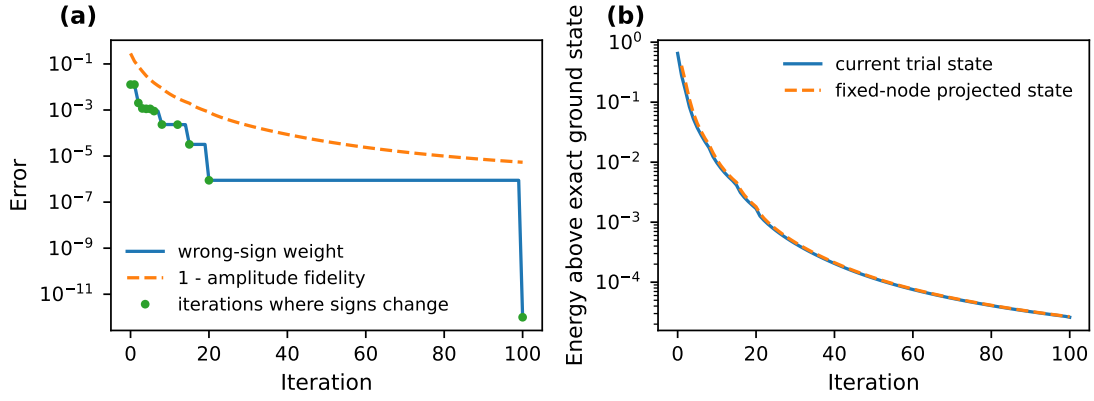


FIG. 3. Exact 4×4 fixed-node/Krylov loop at $J_2/J_1 = 0.5$. (a) Wrong-sign weight w_s and amplitude error $1 - F_a$, with $F_a = |\langle a|a_0 \rangle|^2$. Markers indicate iterations at which the sign pattern changes. (b) Energy above the exact ground state for the current trial state and the fixed-node projected state.

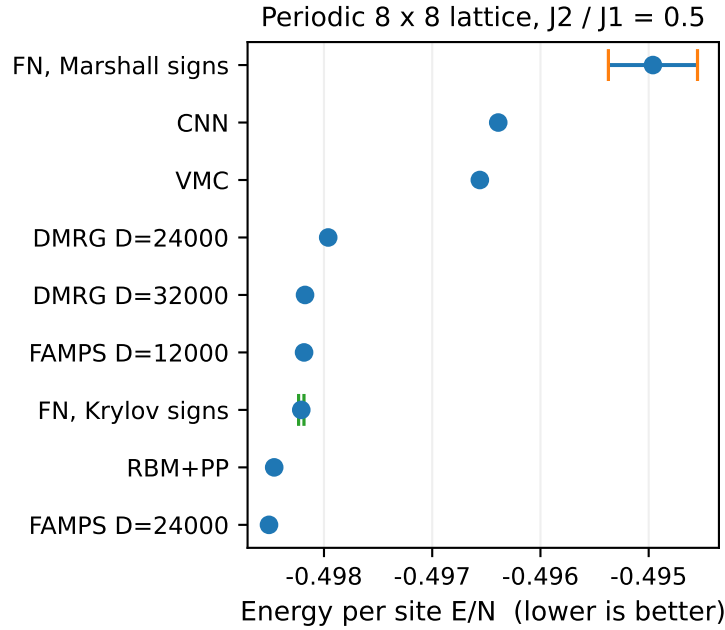


FIG. 4. Energy per site on the periodic 8×8 lattice at $J_2/J_1 = 0.5$. “FN” denotes lattice fixed node. The two FN entries use Marshall signs or the one-step Krylov signs. Short ticks show the two independent walker populations. Literature values are from Ref. [7].

Appendix B: Statistical convention

Exact-diagonalization data are deterministic up to eigensolver precision. For the 6×6 fixed-amplitude com-

parison we use correlation-robust between-chain standard errors. For 8×8 fixed-node data with two independent walker populations, error bars denote half the population difference and are treated only as reproducibility scales.

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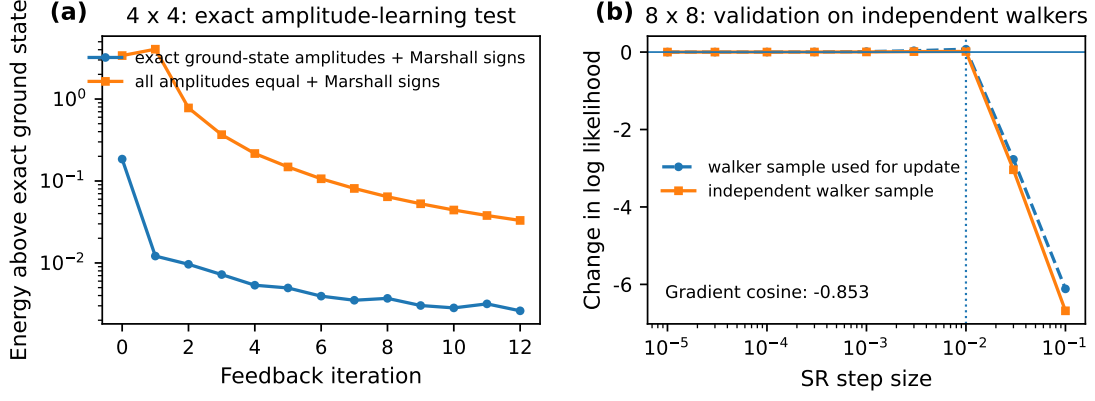


FIG. 5. Learning the positive amplitude. (a) Exact 4×4 test of Eq. (8). The two curves start from the exact target amplitudes or from equal positive amplitudes, in both cases with Marshall signs. (b) Stochastic-reconfiguration (SR) step-size scan on 8×8 : the update is fitted to one fixed-node walker sample and evaluated on an independent sample.

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